

Chapter 1 Introduction

Minghui Shi

Xiamen University, China

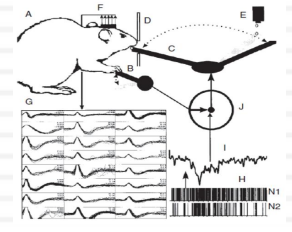
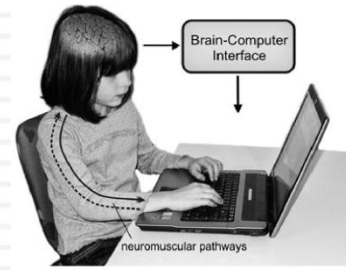
Goal

- Understand BCI
- Overview the content of the whole course
- Arouse interest in BCI

Outline



- Principle of Brain Computer Interface
- Content of this textbook
- Picture & Video



Principle of Brain Computer Interface (BCI)

First Impression on BCI

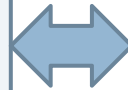
- [Video](#) (19 min) [Chinese]
 - ▣ 2021世界机器人大赛采访高小榕胥红来
- [Video](#) (5 min) [English]
 - ▣ 华盛顿大学的神经工程师Rajesh Rao正在开发脑机接口
- [Video](#) (3 min) [English]
 - ▣ An introduction to neural interfaces
- [Video](#) (26 min) [Chinese]
 - ▣ 李骁健：脑机接口热知识
- [Video](#) (12 min) [Chinese]
 - ▣ 一个视频讲清楚脑机接口技术
- [Video](#) (20 min) [Chinese]
 - ▣ 脑机接口市场可达107亿美元

Principle of Brain Computer Interface

Goal & Outline

1. Learning

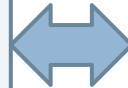
- Basic concepts
- Application values



What is Brain Computer Interface?

2. Understanding

- System framework
- Running mechanism



What are the components in the framework of a BCI system?

How does the framework work?

3. Mastering

- Critical component
- Intention Identification



What is the critical component?

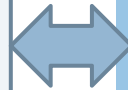
How to identify intention from brain signals?

Principle of Brain Computer Interface

Goal & Outline

1. Learning

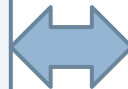
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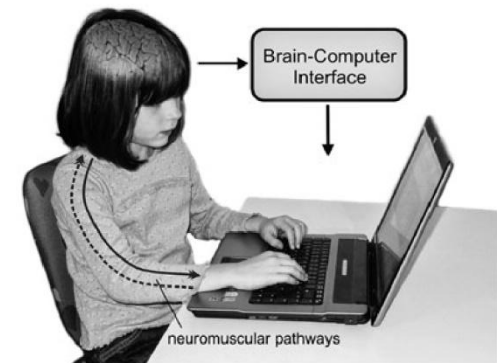
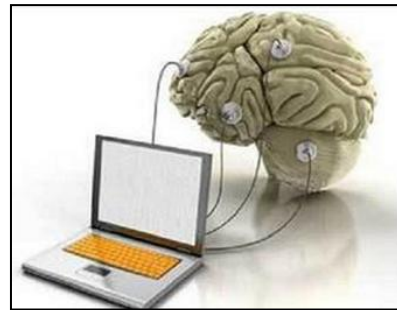
What is the critical component?

How to identify intention from brain signals?

What is Brain Computer Interface?

Question 1:

Are there other ways for us to communicate with a computer?



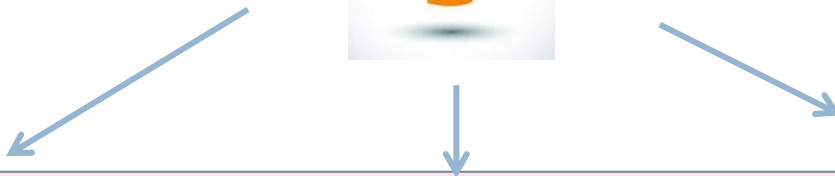
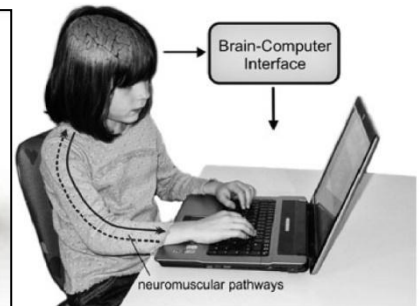
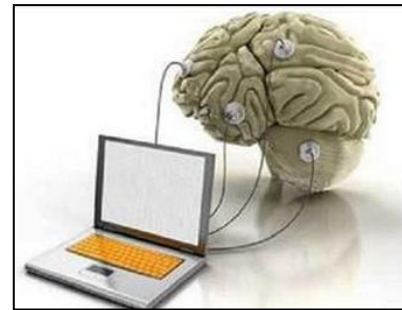
Muscle

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What is Brain Computer Interface?

Question 2:

How can we use our **brain** to control a machine **directly**?



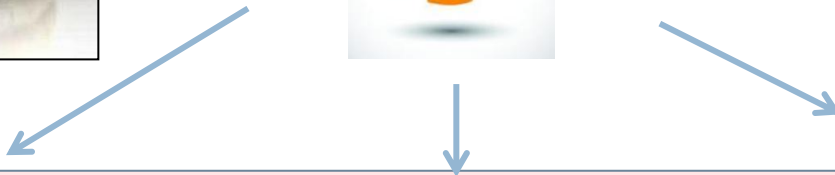
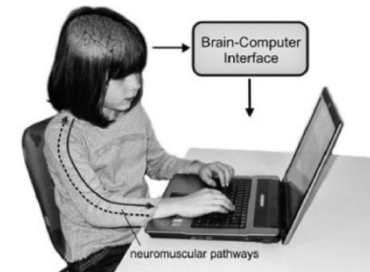
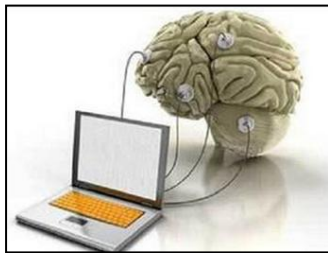
Brain Computer Interface (BCI)



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What is Brain Computer Interface?

- **BCI** is a system that can allow users to communicate with the surrounding directly by their brain.



Brain Computer Interface (BCI)



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What is Brain Computer Interface?

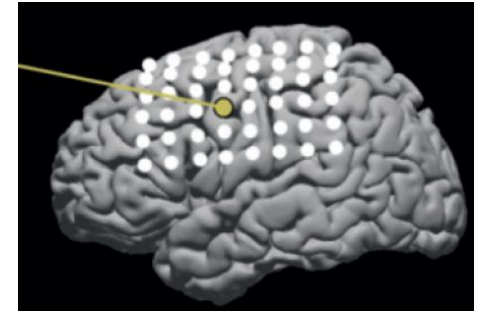
- Three types of BCI:
 - ▣ **Invasive BCI**
 - ▣ **Semi-invasive BCI**
 - ▣ **Non-invasive BCI**

What is Brain Computer Interface?

Invasive BCI



Semi-invasive BCI



Non-invasive BCI



What is Brain Computer Interface?

□ Video

▣ Invasive or Semi-invasive BCI

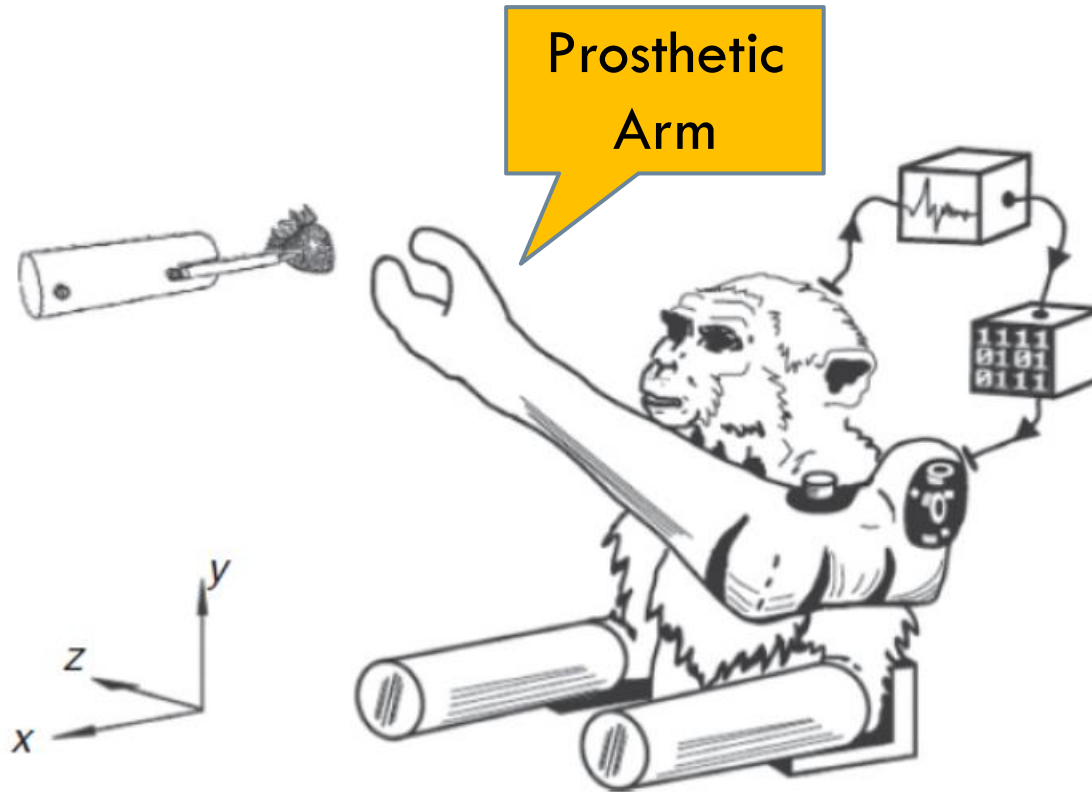
- Control devices and check emails with brain
- Monkey's robotic arm

▣ Non-invasive BCI

- Play a game with brain
- Skeleton: Video 1 Video 2 Video3

What is Brain Computer Interface?

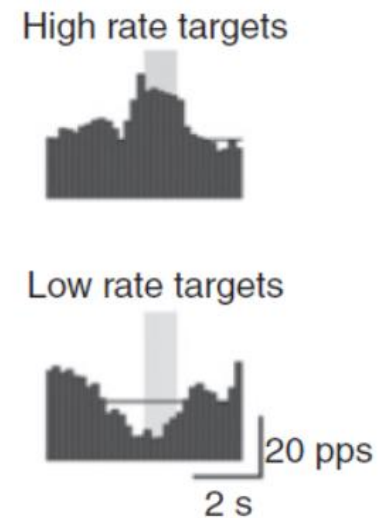
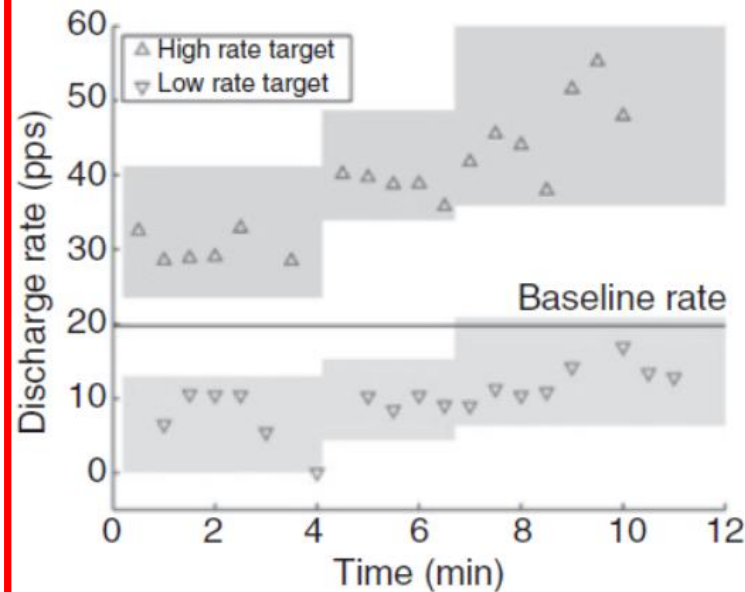
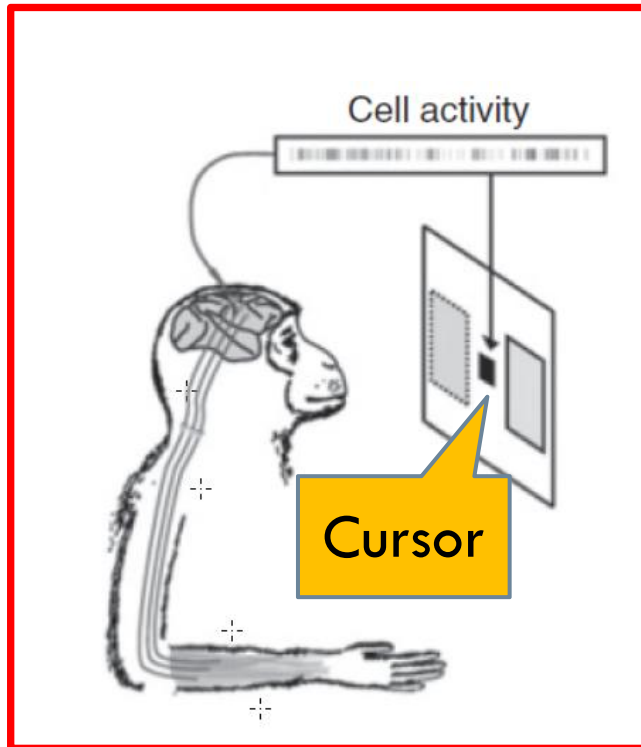
--- BCI Experiments and Applications



BCI Application: BCI Control of a Prosthetic Arm.

What is Brain Computer Interface?

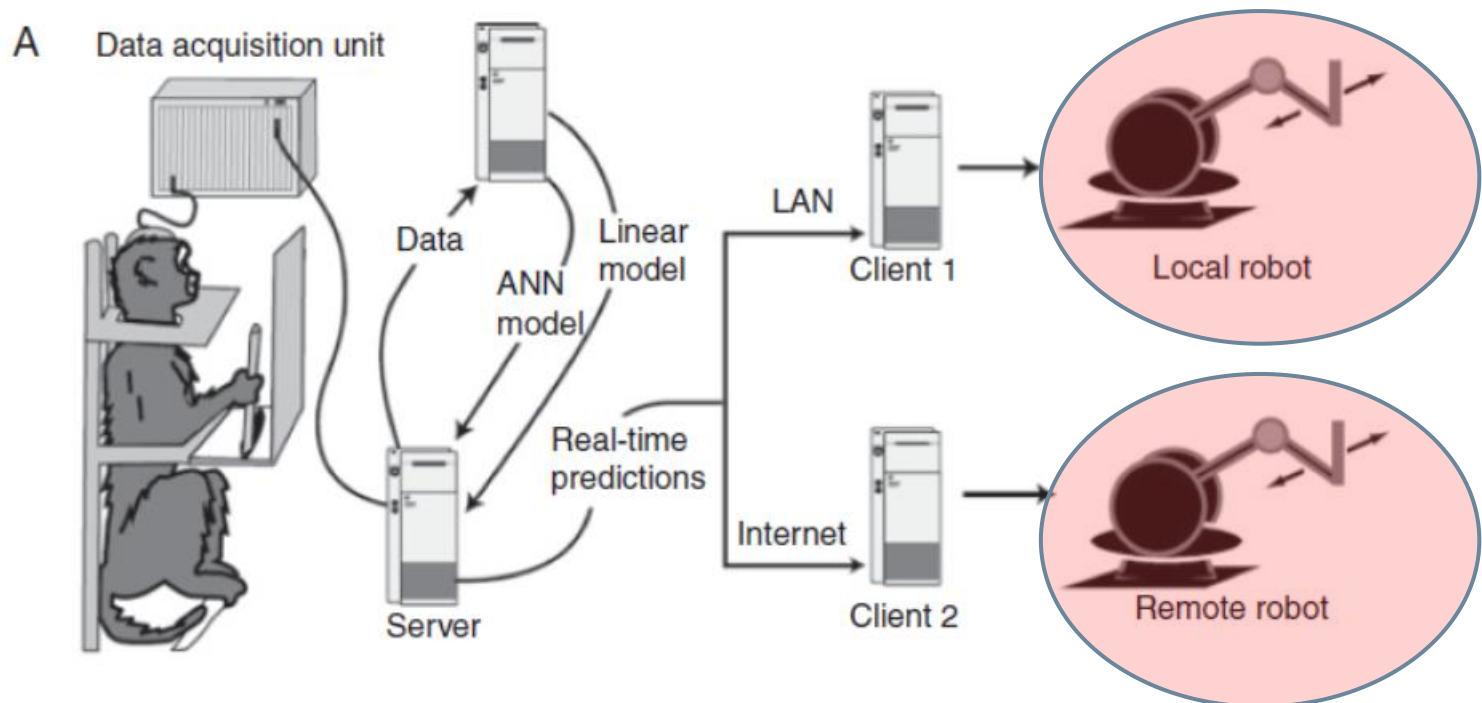
--- BCI Experiments and Applications (Con.)



BCI Applications: BCI Control of a Cursor.

What is Brain Computer Interface?

--- BCI Experiments and Applications (Con.)



BCI Applications: Monkey BCI for Control Tasks.

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What is Brain Computer Interface?

--- BCI Experiments and Applications (Con.)



BCI Applications: for Gaming and Spelling.

What is Brain Computer Interface?

--- BCI Experiments and Applications (Con.)

Rehabilitation



Virtual Reality



BCI Applications: for Rehabilitation and Virtual Reality.

What is Brain Computer Interface?

--- BCI Experiments and Applications (Con.)

Intelligent Machine Dogs



Dial System



BCI Applications: for Intelligent Machine Dogs and Dial System.

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What is Brain Computer Interface?

--- BCI Experiments and Applications (Con.)

Cursor Control

Wheel Chair Control



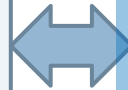
BCI Applications: for **Cursor Control** and **Wheel Chair Control**.

Principle of Brain Computer Interface

Goal & Outline

1. Learning

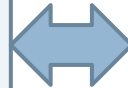
- Basic concepts
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What is Brain Computer Interface?

2. Understanding

- System framework
- Running mechanism



What are the components in the framework of a BCI system?

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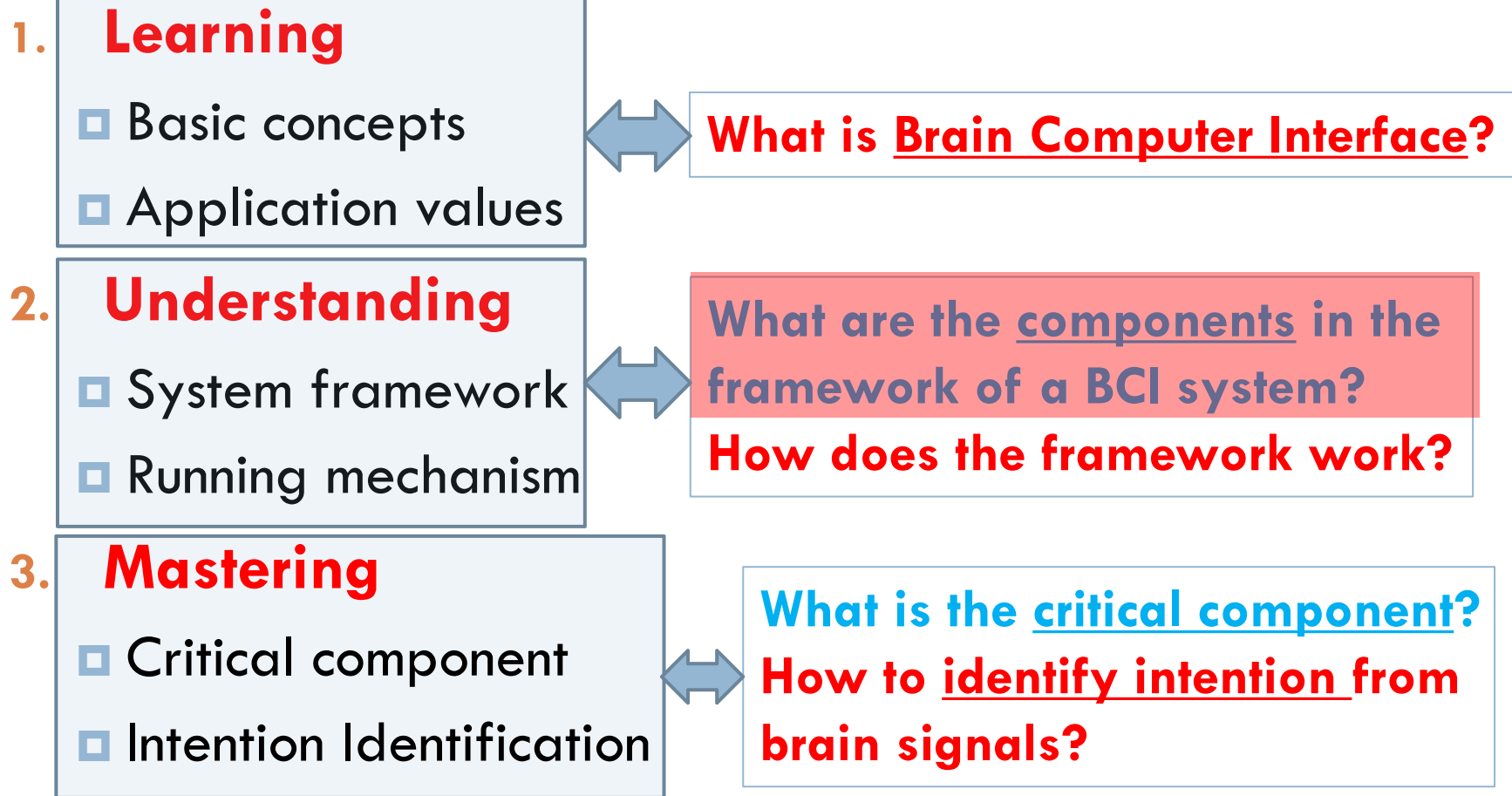


What is the critical component?

How to identify intention from brain signals?

Principle of Brain Computer Interface

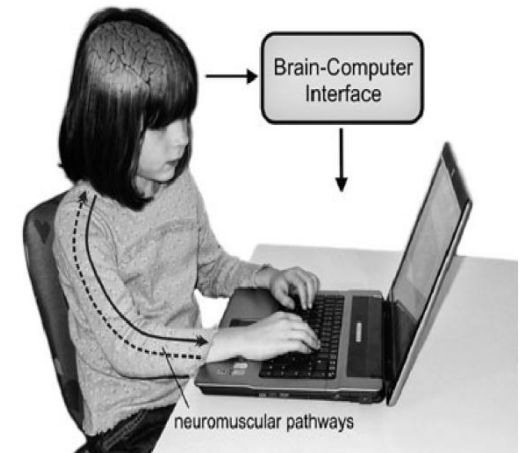
Goal & Outline



What are the components in the framework of a BCI system?

□ Four components in a BCI system:

1. **Recording**
2. **Identifying**
3. **Transforming**
4. **Responding**



Recording

Identifying

Preprocessing
Feature Extraction
Classification

Transforming

Responding

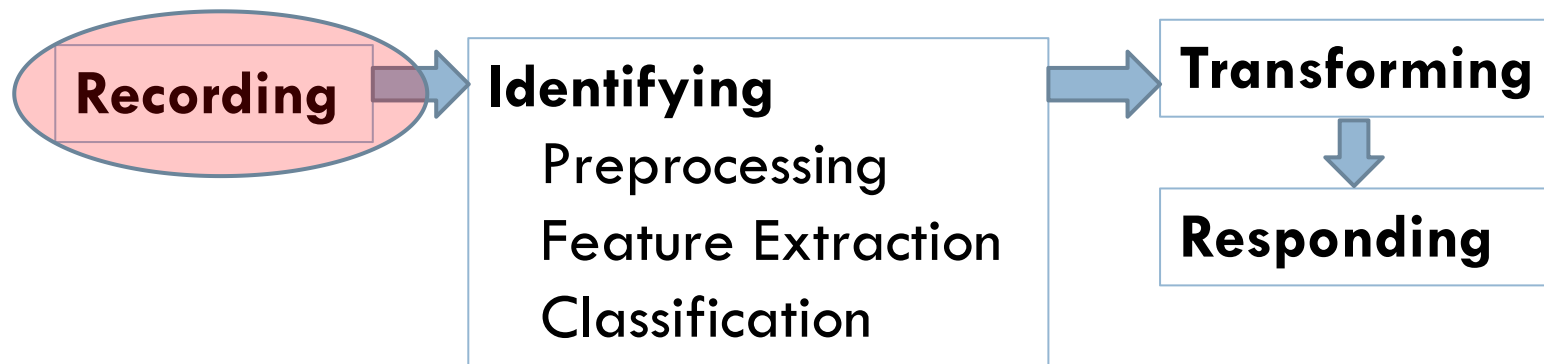
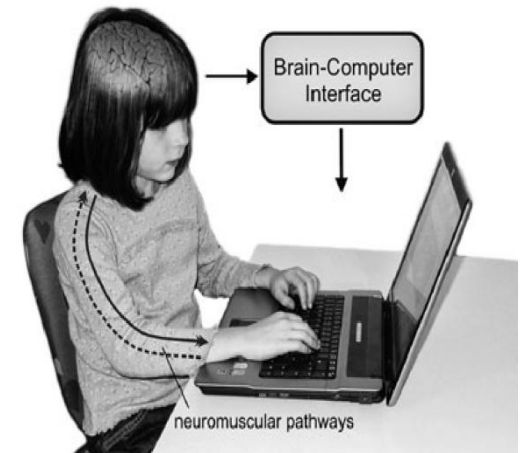
Framework of a BCI system

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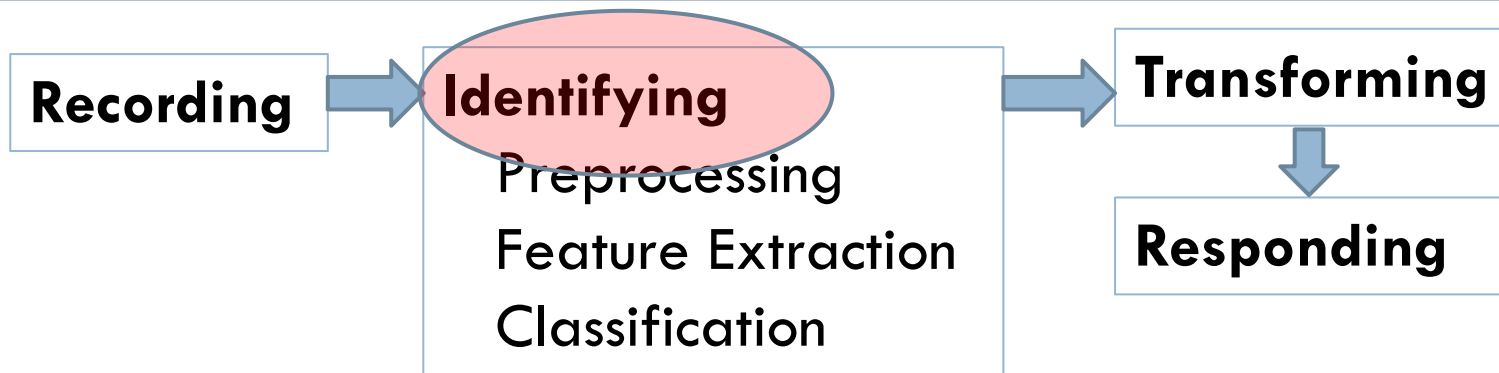
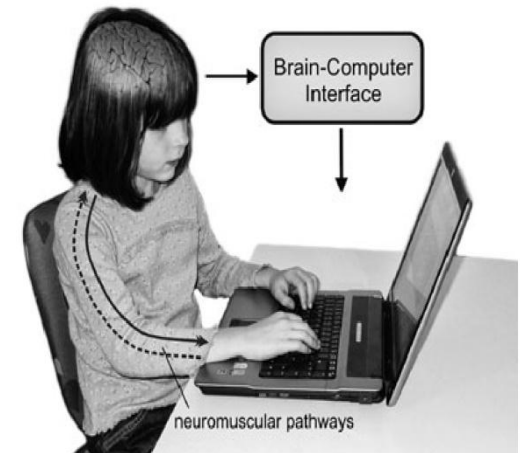
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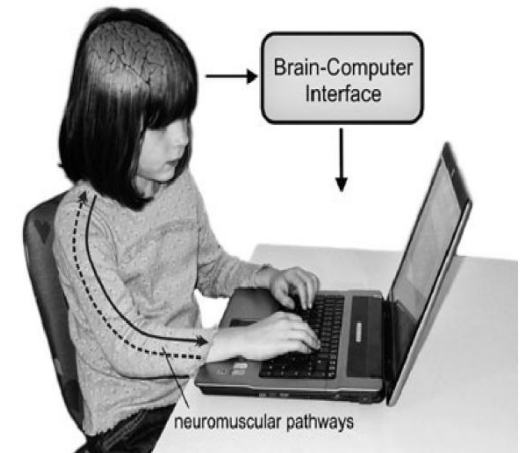
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Recording

Identifying

Preprocessing
Feature Extraction
Classification

Transforming

Responding

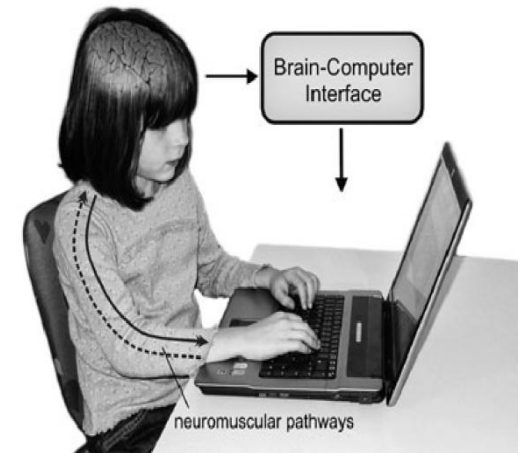
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Framework of a BCI system

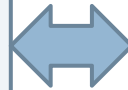
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Principle of Brain Computer Interface

Goal & Outline

1. Learning

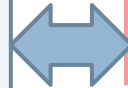
- Basic concepts
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What is Brain Computer Interface?

2. Understanding

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What are the components in the framework of a BCI system?

How does the framework work?

3. Mastering

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What is the critical component?

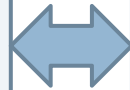
How to identify intention from brain signals?

Principle of Brain Computer Interface

Goal & Outline

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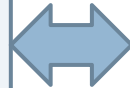
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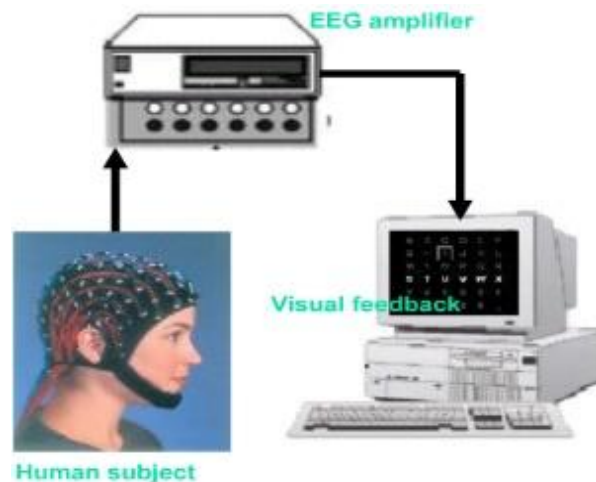
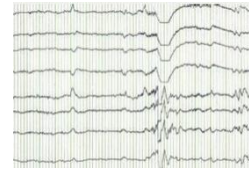
What is the critical component?

How to identify intention from brain signals?

How does the framework work?

□ 1. Recording

- ▣ Record the brain signals.
- ▣ Feed the recorded brain signals into a computer.



How does the framework work?

□ Recording devices



How does the framework work?

□ 1. Recording

- Record the brain signals.

□ 2. Identifying

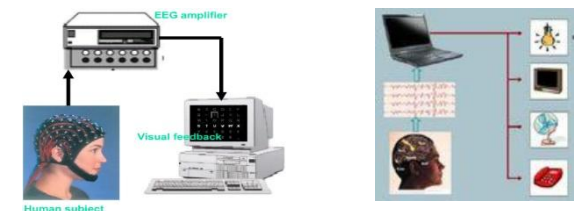
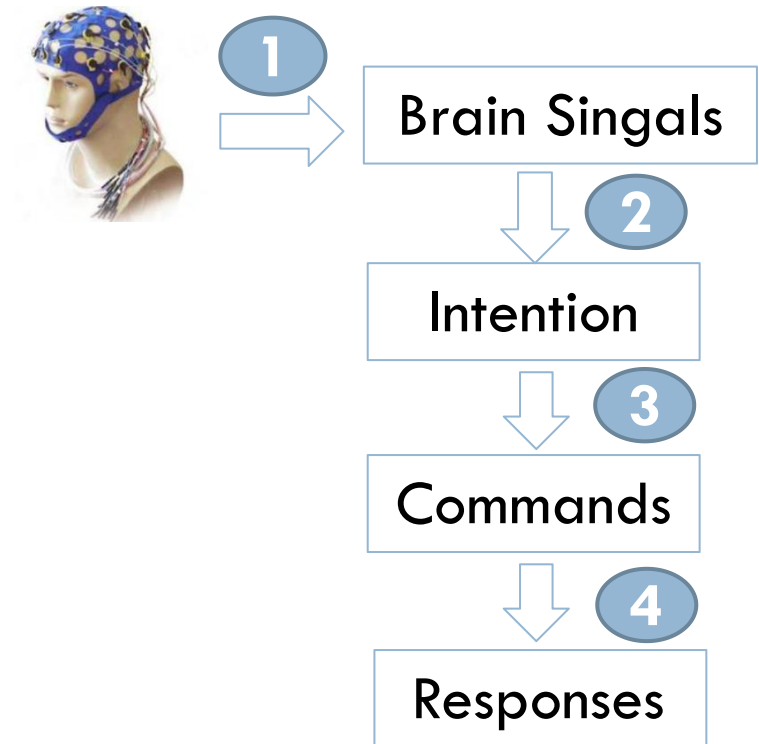
- Identify user's intention, by analyzing the recorded brain signals.

□ 3. Transforming

- Transform the intention into corresponding commands.

□ 4. Responding

- Make responses.

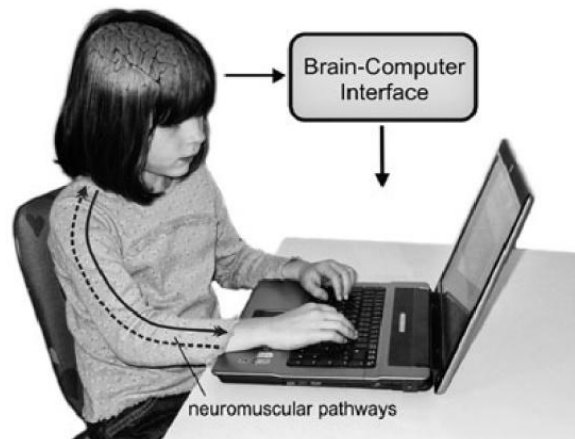


How does the framework work?

--- First Example

□ Intention:

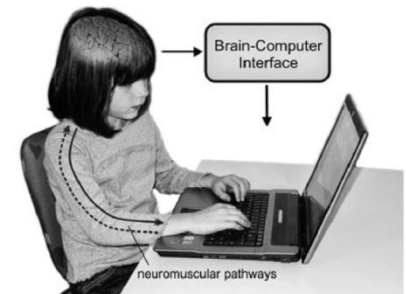
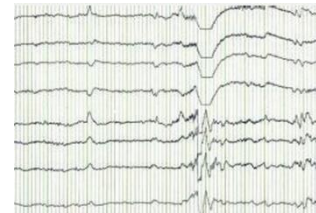
- Input the word “**Sky**” into the computer by BCI.



How does the framework work?

--- First Example (Con.)

- The BCI system will undergo four steps:
- **1. Recording**
 - ▣ Record the girl's *brain signals*.
- **2. Identifying**
 - ▣ Identify the girl's *intention*: type three letters ('**S**', '**K**', '**Y**').
- **3. Transforming**
 - ▣ Transform the *intention* into *commands*: present the letters ('**S**', '**K**', '**Y**').
- **4. Responding**
 - ▣ Make *responses*, i.e., present the letters ('**S**', '**K**', '**Y**') on the screen.

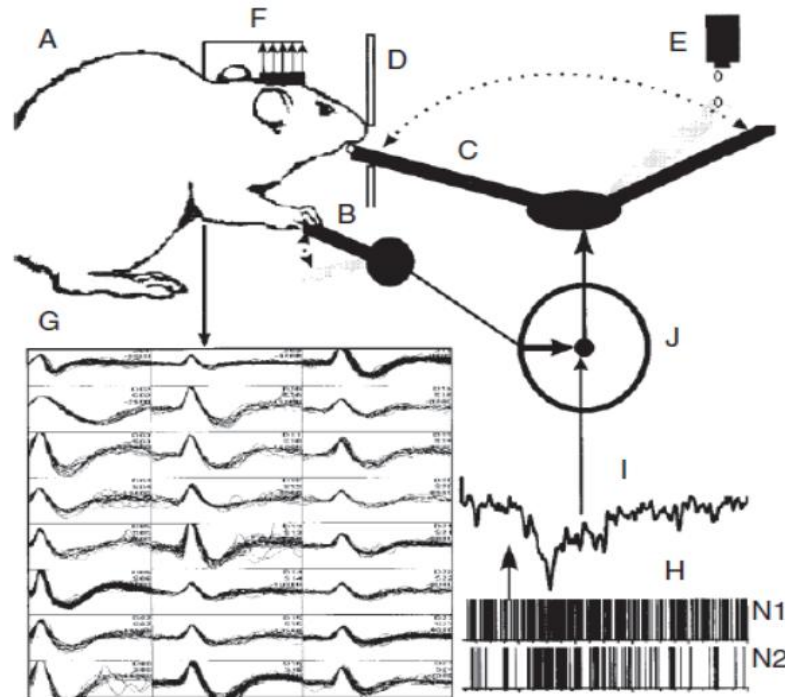


How does the framework work?

--- Second Example

□ Intention:

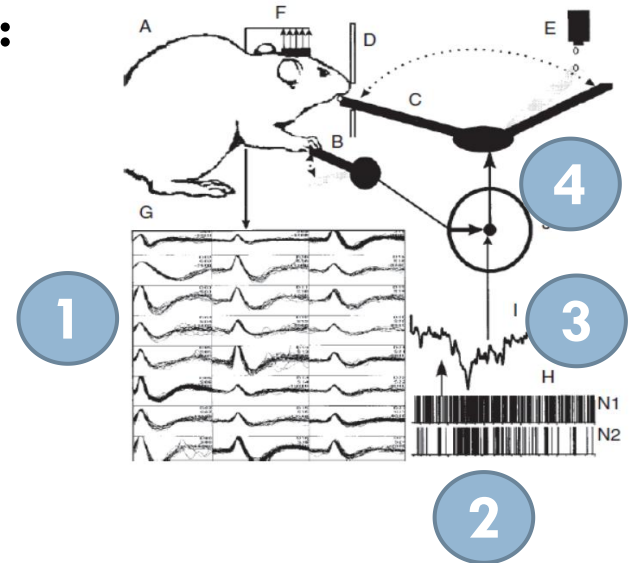
- Raise the control part of the level.



How does the framework work?

--- Second Example (Con.)

- The BCI system will undergo four steps:
- **1. Recording**
 - ▣ Record the rat's *brain signals*.
- **2. Identifying**
 - ▣ Identify the *intention*: raise the level.
- **3. Transforming**
 - ▣ Transform the intention into *commands*.
- **4. Responding**
 - ▣ Make *response*: raise the level.

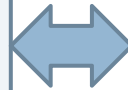


Principle of Brain Computer Interface

Goal & Outline

1. Learning

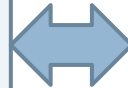
- Basic concepts
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What is Brain Computer Interface?

2. Understanding

- System framework
- Running mechanism



What are the components in the framework of a BCI system?

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What is the critical component?

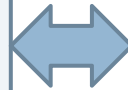
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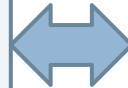
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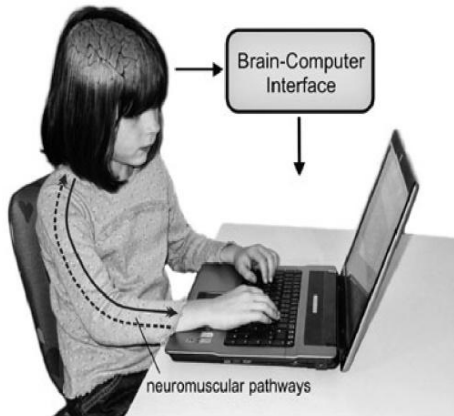
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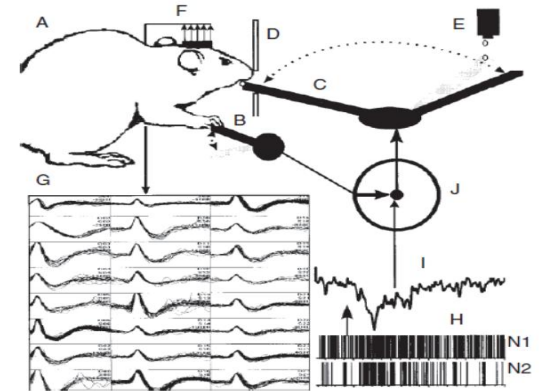


What is the critical component?
How to identify intention from brain signals?

What is the critical component?



1. Recording
2. Identifying
3. Transforming
4. Responding



Recording

Identifying

Preprocessing
Feature Extraction
Classification

Transforming

Responding

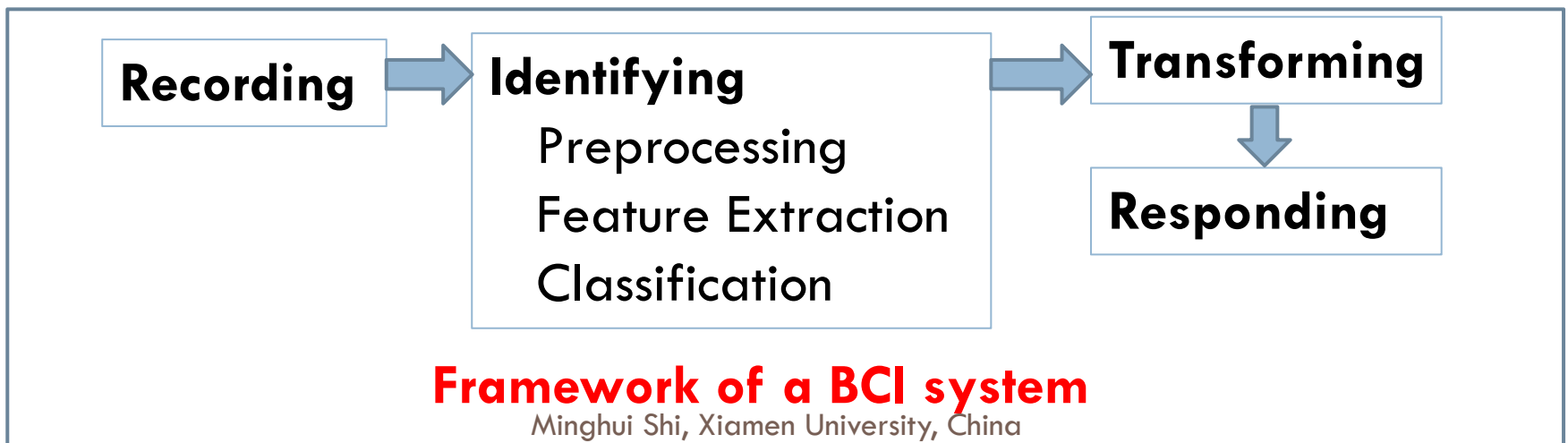
Framework of a BCI system

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What is the critical component?

□ Which one is the critical one?

1. **Recording**
2. **Identifying**
3. **Transforming**
4. **Responding**



What is the critical component?

□ Which one is the critical one?

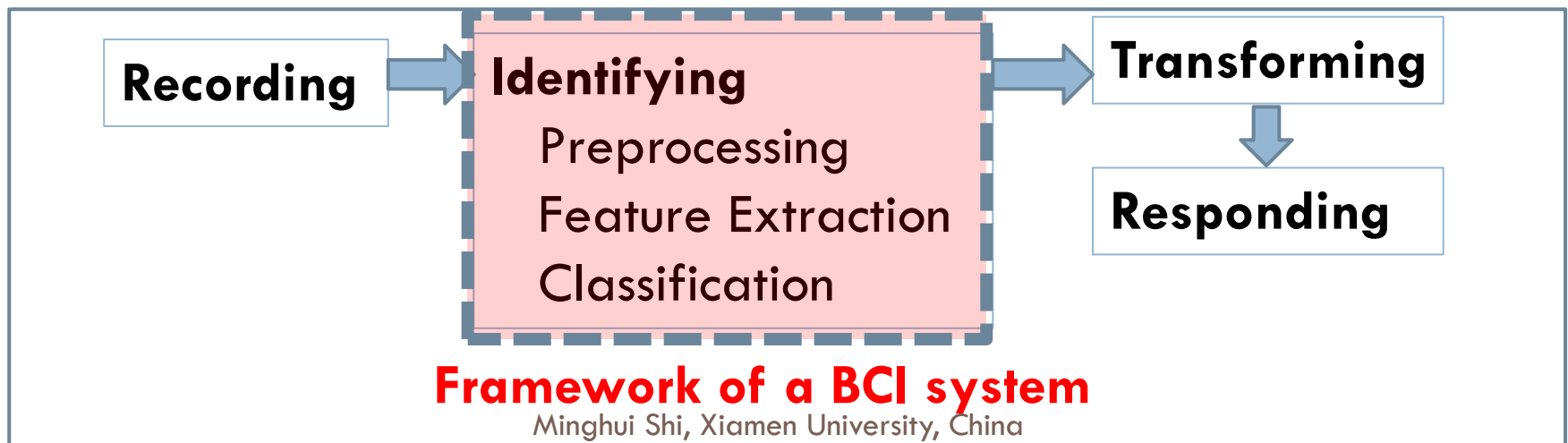
1. **Recording**

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Critical !

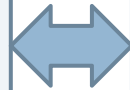


Principle of Brain Computer Interface

Goal & Outline

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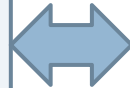
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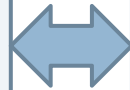
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How to identify intention from brain signals?

Principle of Brain Computer Interface

Goal & Outline

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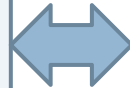
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What is the critical component?
How to identify intention from brain signals?

How to identify intention from brain signals? --- Example



**Particular
Intentions**



**Desire
Responses**

imagined movements:

feet



right hand



left hand



character actions:



run forward



turn right



turn left

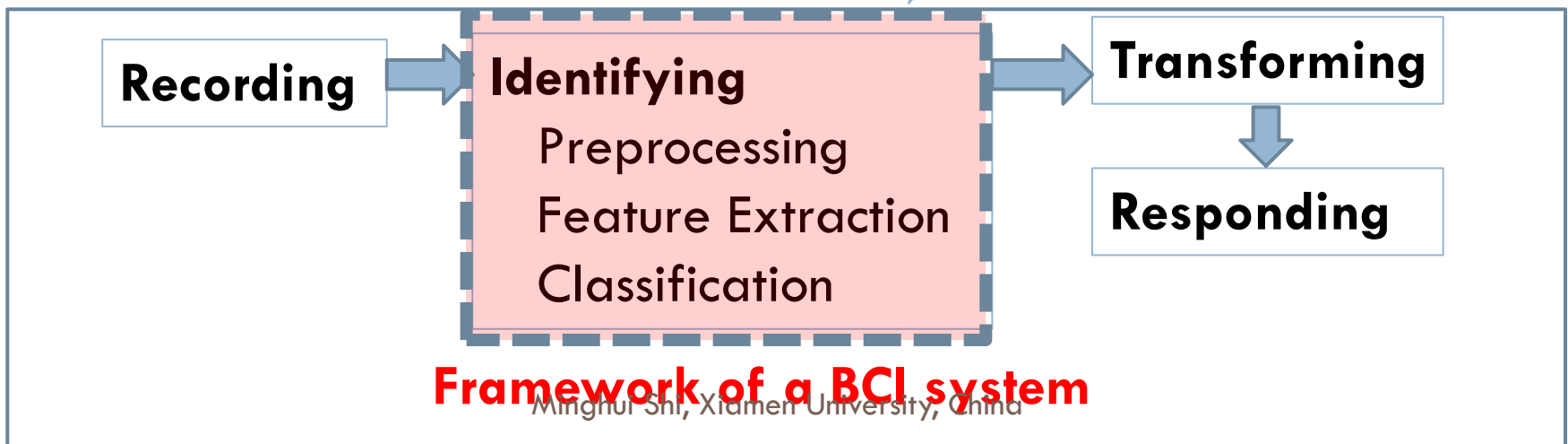
How to identify intention from brain signals?



□ *Three steps* to analyze brain signals:

1. **Preprocessing**
2. **Feature extraction**
3. **Classification**

Critical !

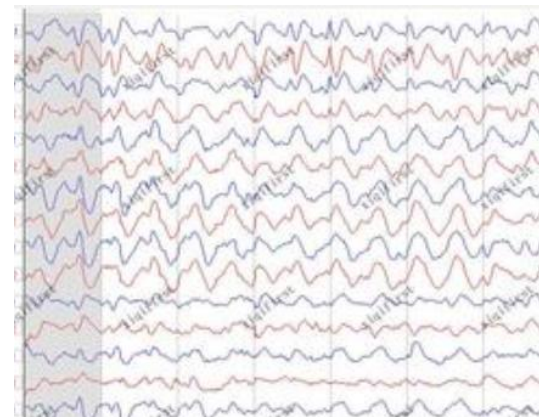
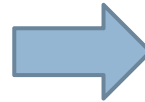
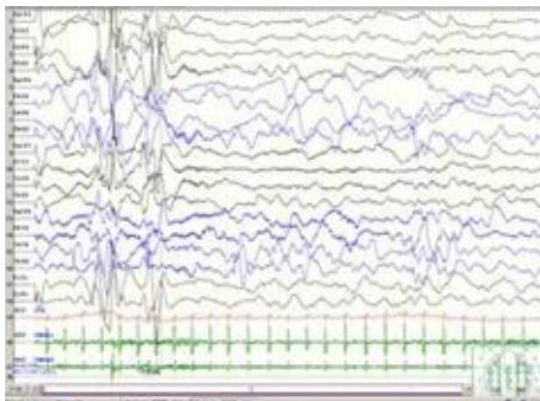


How to identify intention from brain signals?



□ 1. Preprocessing

- ▣ Enhance *signal-noise-rate* (**SNR**).
 - **Remove** *noise* and *artifacts*.
 - **Enhance** the *components* that are associated with the subject's intention.

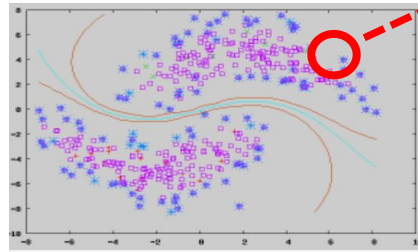


How to identify intention from brain signals?



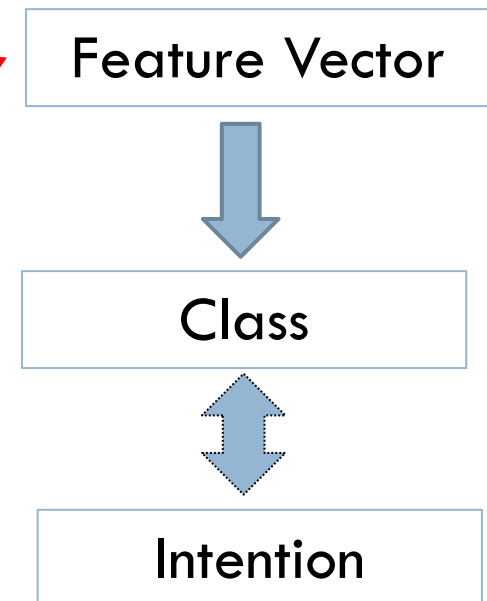
□ 2 Feature Extraction

- Find the *features* hidden in the brain signals.



□ 3 Classification

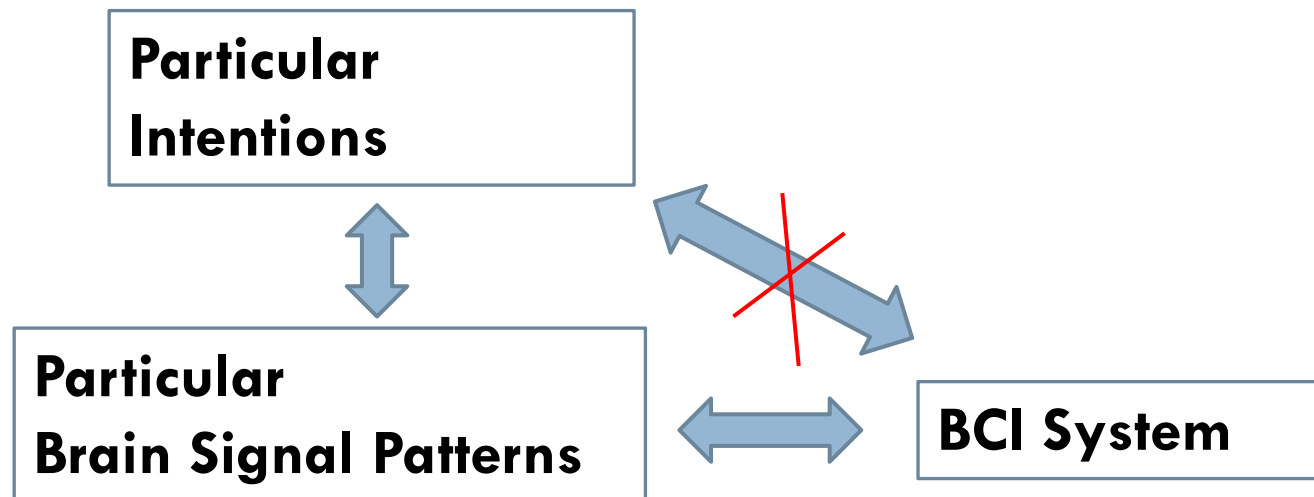
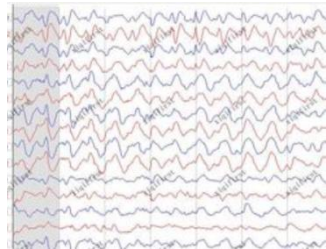
- find a *model* that can correctly classify the feature vectors.



How to identify intention from brain signals?



Is BCI system a **mind-reading** system?

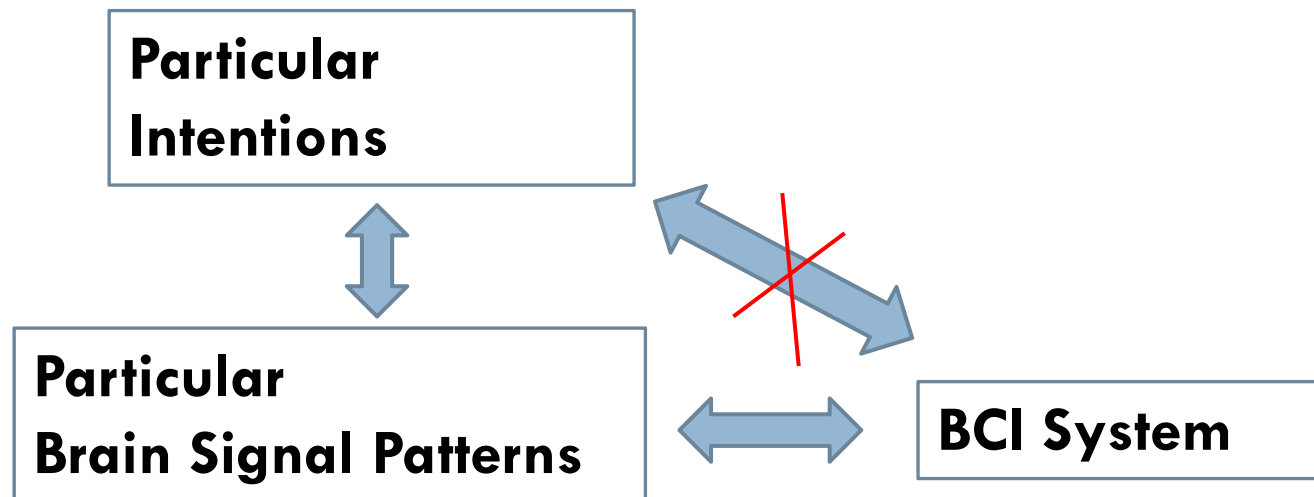
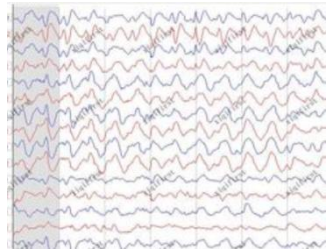


How to identify intention from brain signals?



Is BCI system a **mind-reading** system?

BCI system is a **pattern-reading** system.



How to identify intention from brain signals?



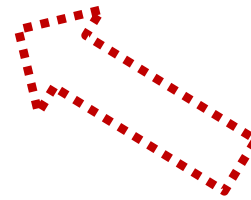
BCI system is not a **mind-reading** system

BCI system is a **pattern-reading** system.

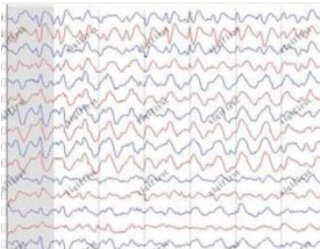
Particular Intentions



Particular Brain Signal Patterns

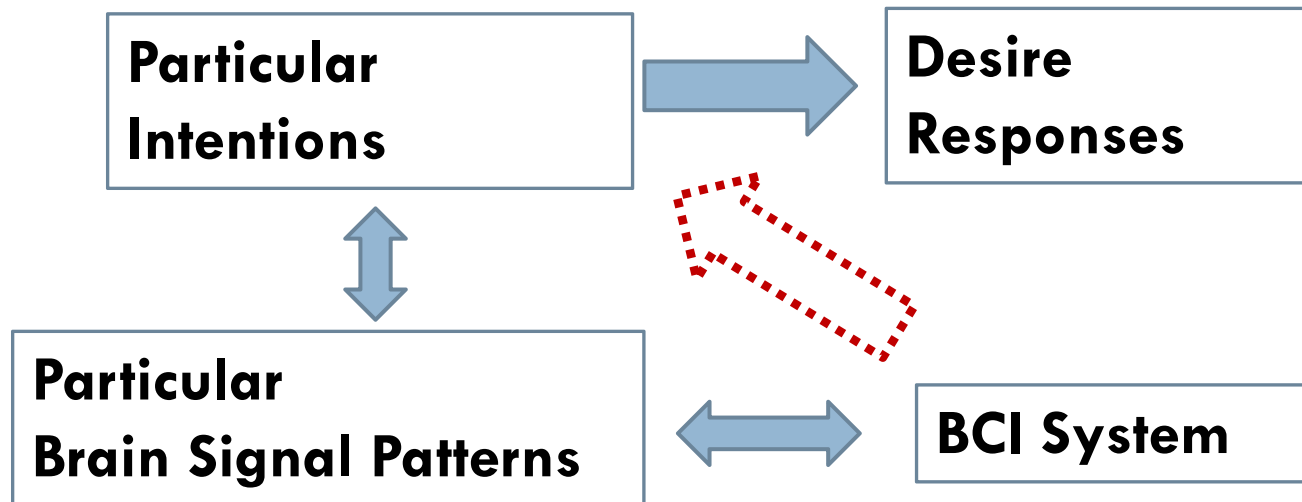


BCI System



Note: a BCI system is not a **mind-reading** system
but a **pattern-reading** system.

How to identify intention from brain signals?



Note: a BCI system is not a **mind-reading** system
but a **pattern-reading** system.

How to identify intention from brain signals? --- Example (Review)



**Particular
Intentions**



**Desire
Responses**

imagined movements:

feet



right hand



left hand



character actions:



run forward



turn right



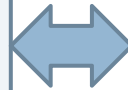
turn left

Principle of Brain Computer Interface

Goal & Outline

1. Learning

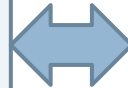
- Basic concepts
- Application values



What is Brain Computer Interface?

2. Understanding

- System framework
- Running mechanism



What are the components in the framework of a BCI system?
How does the framework work?

3. Mastering

- Critical component
- Intention Identification



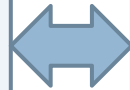
What is the critical component?
How to identify intention from brain signals?

Principle of Brain Computer Interface

Summary

1. Learning

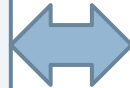
- Basic concepts
- Application values



What is Brain Computer Interface?

2. Understanding

- System framework
- Running mechanism



What are the components in the framework of a BCI system?
How does the framework work?

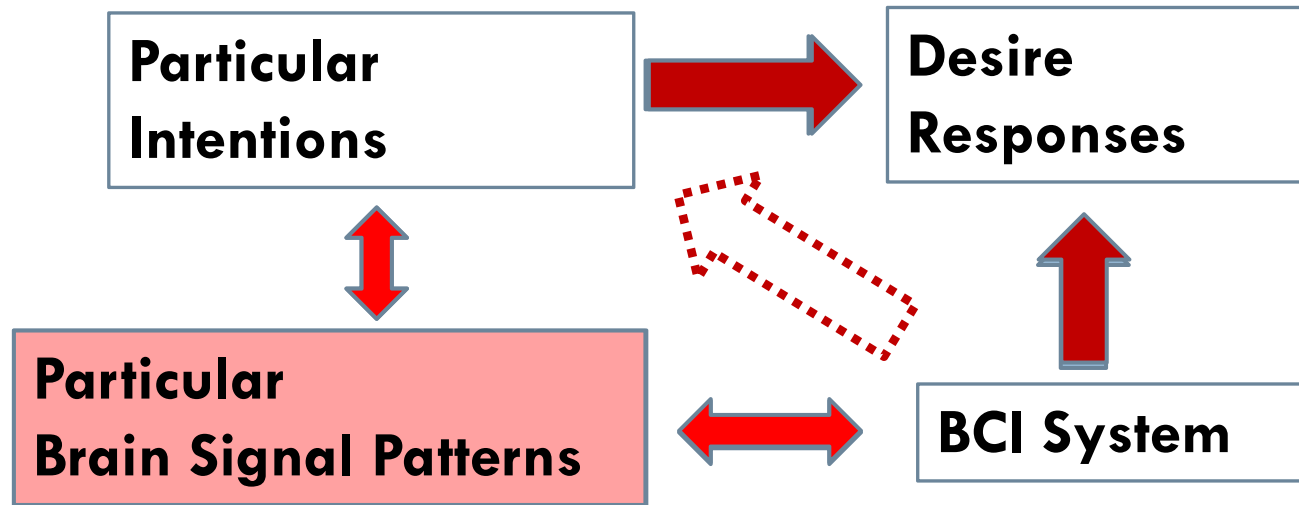
3. Mastering

- Critical component
- Intention Identification



What is the critical component?
How to identify intention from brain signals?

Principle of Brain Computer Interface



Note: a BCI system is not a **mind-reading** system
but a **pattern-reading** system.

Exercise

□ Exercise 1

- ▣ What are the components in a BCI system?
Please describe the purposes of these components.

□ Exercise 2

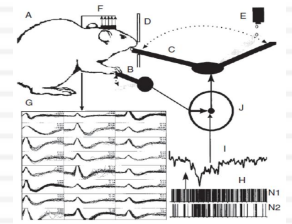
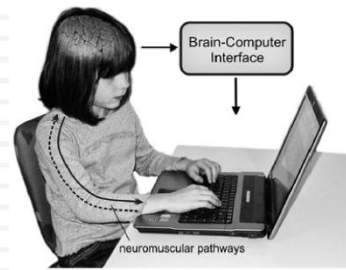
- ▣ Please explain why a BCI system is not a mind-reading system but a pattern-reading system?

□ Exercise 3

- ▣ Describe the BCI principle in detail with your understanding.

Outline

- Principle of Brain Computer Interface
- Content of this textbook
- Picture & Video



Content of this textbook

Processing Stages of a BCI System

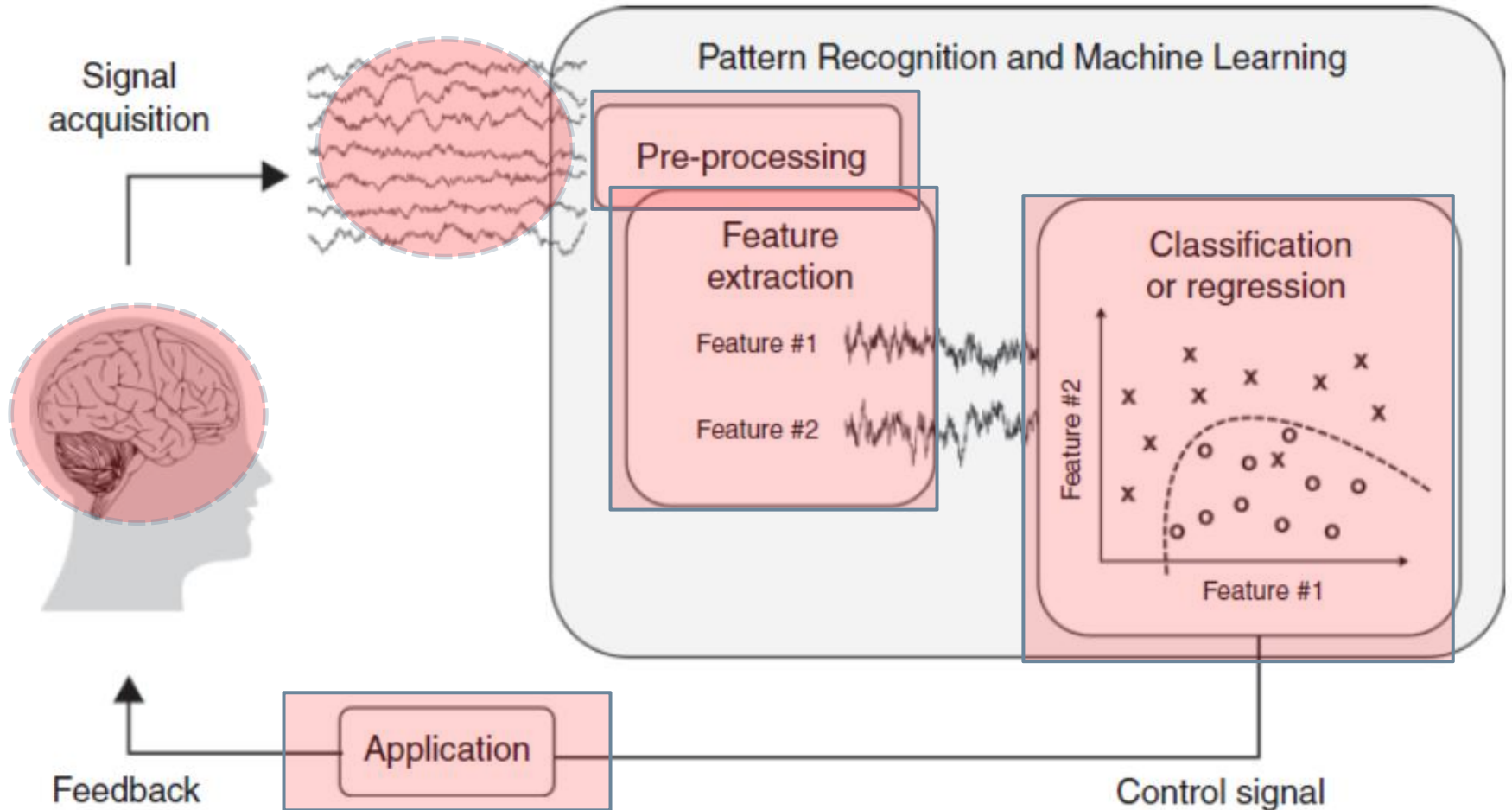


Figure 1.1. **Basic components of a brain-computer interface (BCI).**

Content & Topics

- This course focuses on the topic of **brain-computer interface (BCI)**, an emerging research on human-machine interaction in recent years.
- **The basic content include:**
 - basic neuroscience *background*
 - recording and analyzing **methods** for brain signals
 - fundamental *principle* of BCI
 - various BCI *applications*
 - *ethical* and *social* issues ensued from BCI

Content & Topics

- **1. Introduction**
- **Part I Background**
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**

Content & Topics

- **1. Introduction**
- **Part I Background**
 - ▣ **2. Basic Neuroscience**
 - ▣ **3. Recording and Stimulating the Brain**
 - ▣ **4. Signal Processing**
 - ▣ **5. Machine Learning**
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**

Content & Topics

- **1. Introduction**
- **Part I Background**
- **Part II Putting It All Together**
 - ▣ **6. Building a BCI**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**

Content & Topics

- **1. Introduction**
- **Part I Background**
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
 - ▣ **7. Invasive BCIs**
 - ▣ **8. Semi-Invasive BCIs**
 - ▣ **9. Noninvasive BCIs**
 - ▣ **10. BCIs that Stimulate**
 - ▣ **11. Bidirectional and Recurrent BCIs**
- **Part IV Applications and Ethics**

Content & Topics

- **1. Introduction**
- **Part I Background**
- **Part II Putting It All Together**
- **Part III Major Types of BCIs**
- **Part IV Applications and Ethics**
 - ▣ **12. Applications of BCIs**
 - ▣ **13. Ethics of Brain-Computer Interface**

Detail of Content

- The book is organized as follows:
- **Chapters 1 through 5** of the book provide the necessary background in neuroscience and quantitative techniques to understand the terminology and methods used in building BCIs.
- In **Chapter 6**, we begin our journey into the world of BCIs by learning about the basic components that go into building a BCI.

- The next part of the book introduces the reader to the three major types of BCIs classified according to degree of invasiveness.
- **Chapter 7** describes invasive BCIs, which utilize devices implanted inside the brain.
- **Chapter 8** describes semi-invasive BCIs, which are based on nerve signals or devices implanted on the surface of the brain.
- **Chapter 9** covers non-invasive BCIs such as those that record electrical signals from the scalp (EEG).

- **Chapter 10** reviews BCIs that stimulate the brain in order to, for example, restore lost sensory or motor function.
- **Chapter 11** introduces the most general type of BCIs, namely, BCIs that both record from and stimulate the brain. In each case, examples of classic experiments as well as the state-of-the-art technologies (circa 2013) are presented.
- **Chapter 12** reviews some of the major applications of BCIs.
- **Chapter 13** considers the ethical issues pertaining to the development and use of BCI technology.

- We conclude in **Chapter 14** with a summary of some of the limitations of present-day BCIs and speculate on the future of the field.
- The book also includes an **Appendix** that provides basic mathematical background in linear algebra and probability theory useful for understanding and implementing BCIs.

Roadmap of the content

Background of BCI (Chapter 1-5)



Types & Brain Responses (Chapter 6)



Major Types of BCI

(Chapter 7 invasive BCI)
(Chapter 8 semi-invasive BCI)
(Chapter 9 non-invasive BCI)



Major Types of BCI

BCIs that stimulate brain
(Chapter 10)



Linear Algebra & Probability Theory
(Appendix)



Limitations and Future of BCI
(Chapter 14)



Ethical Issues of BCI (Chapter 13)



Applications of BCI (Chapter 12)

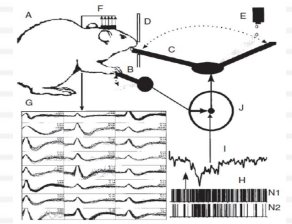
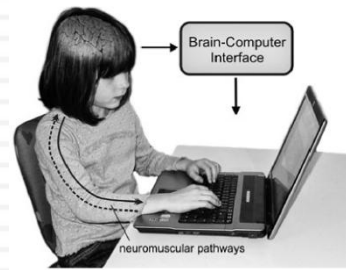


Major Types of BCI

BCIs that both record from and
stimulate the brain (Chapter 11)

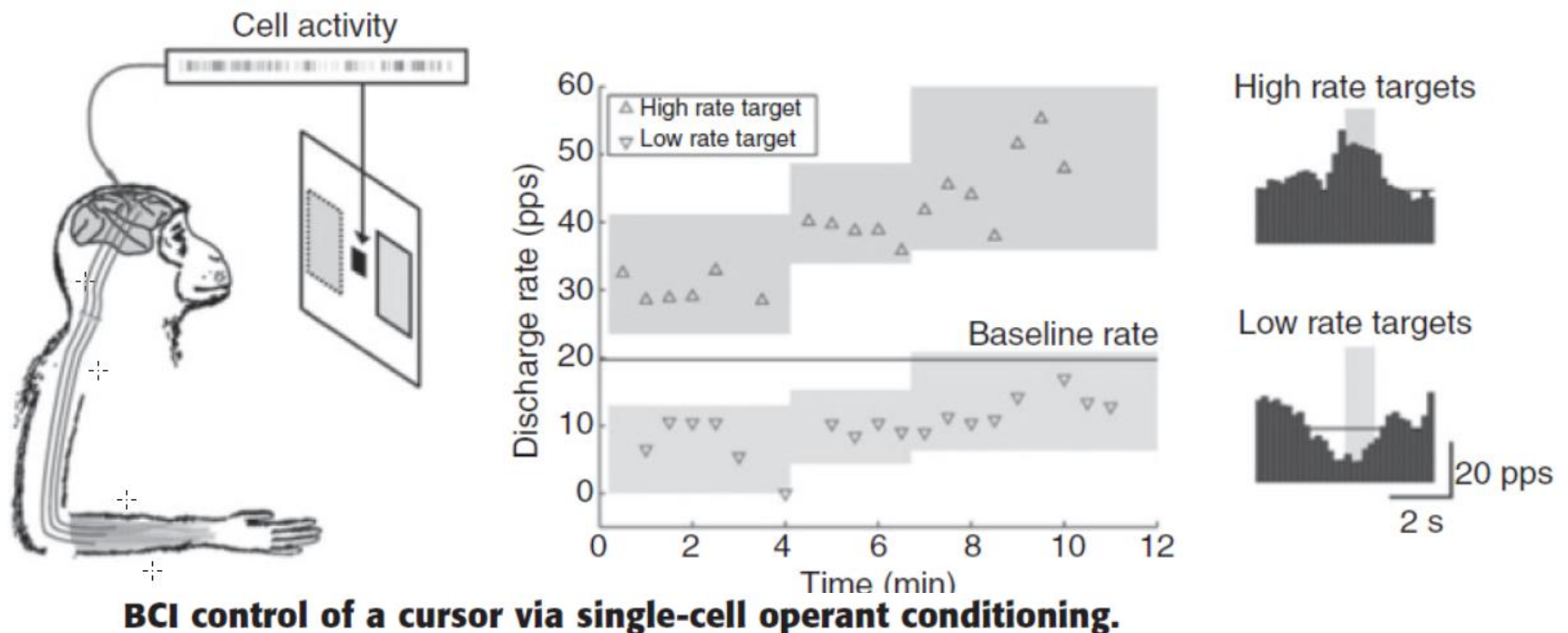
Outline

- Principle of Brain Computer Interface
- Content of this textbook
- Picture & Video

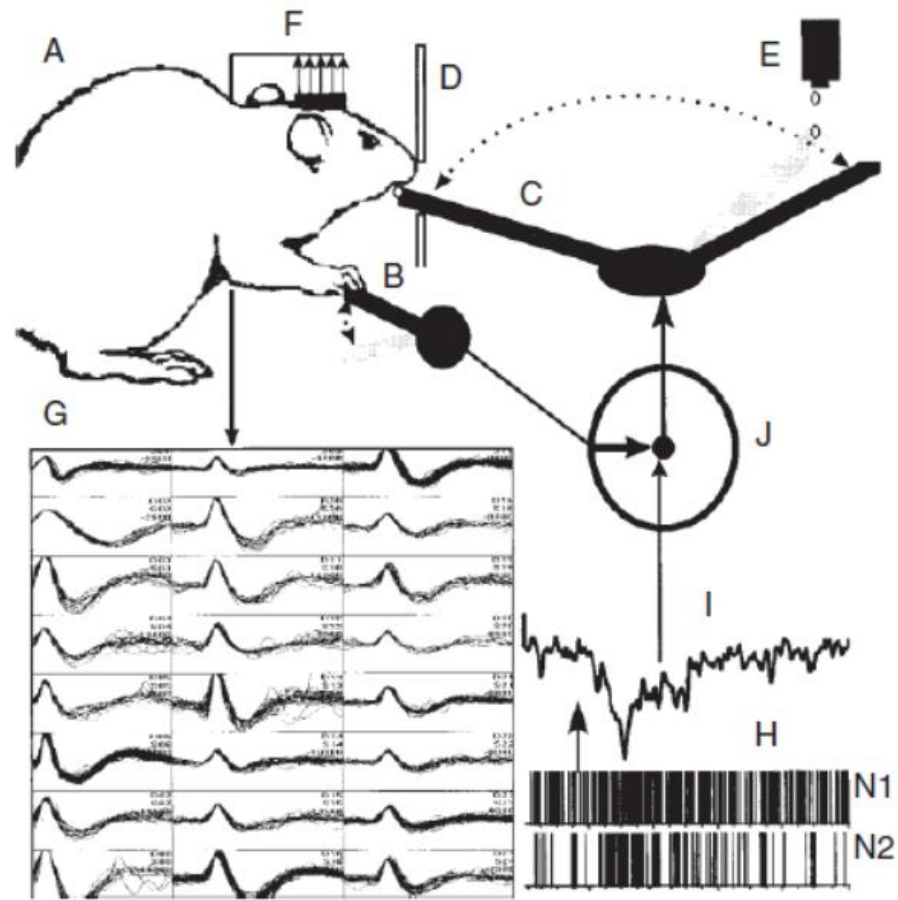


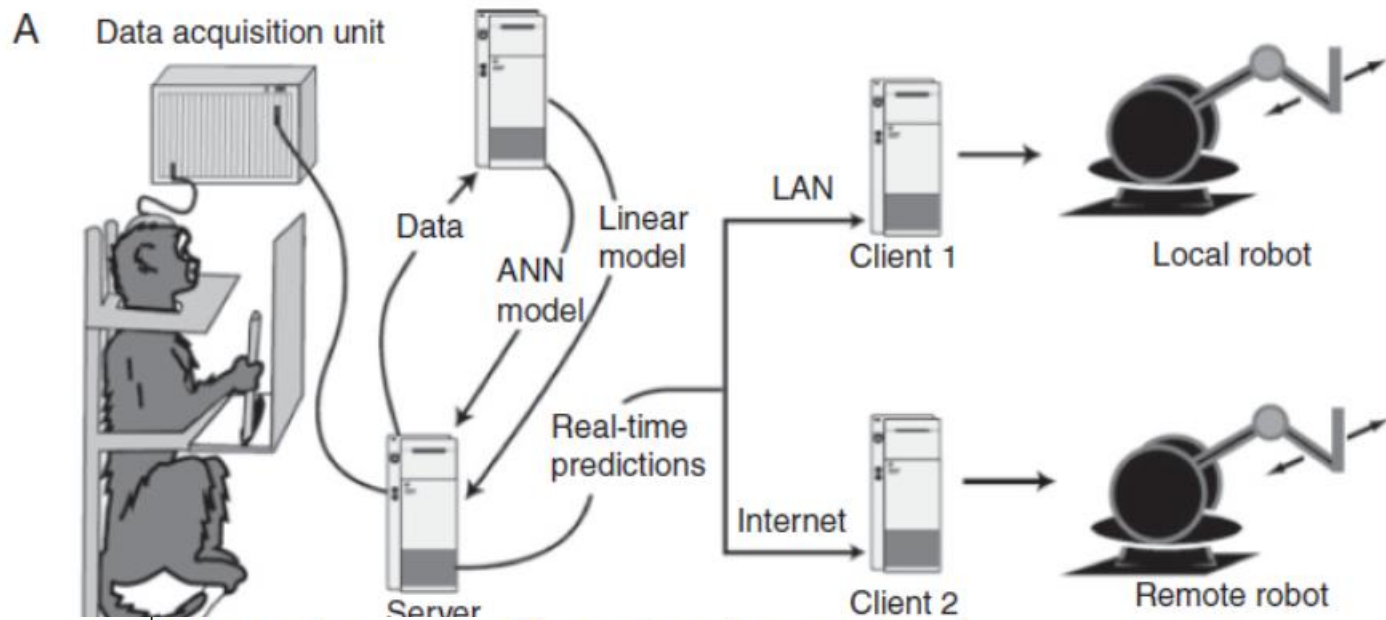
Pictures

BCI Pictures From Textbook

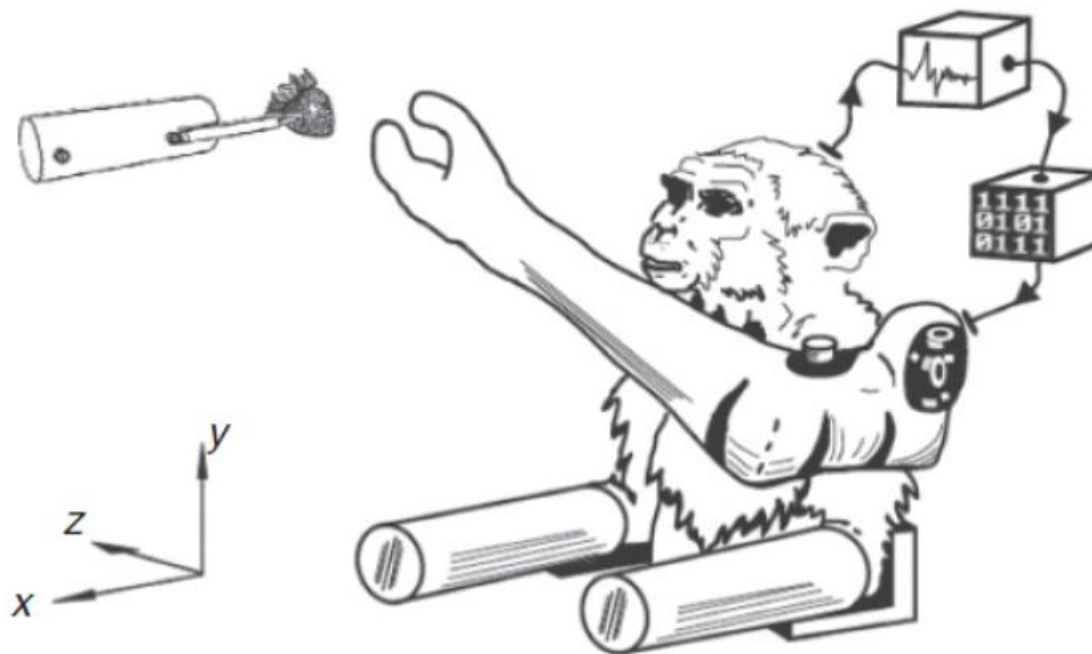


Invasive BCI in rats.

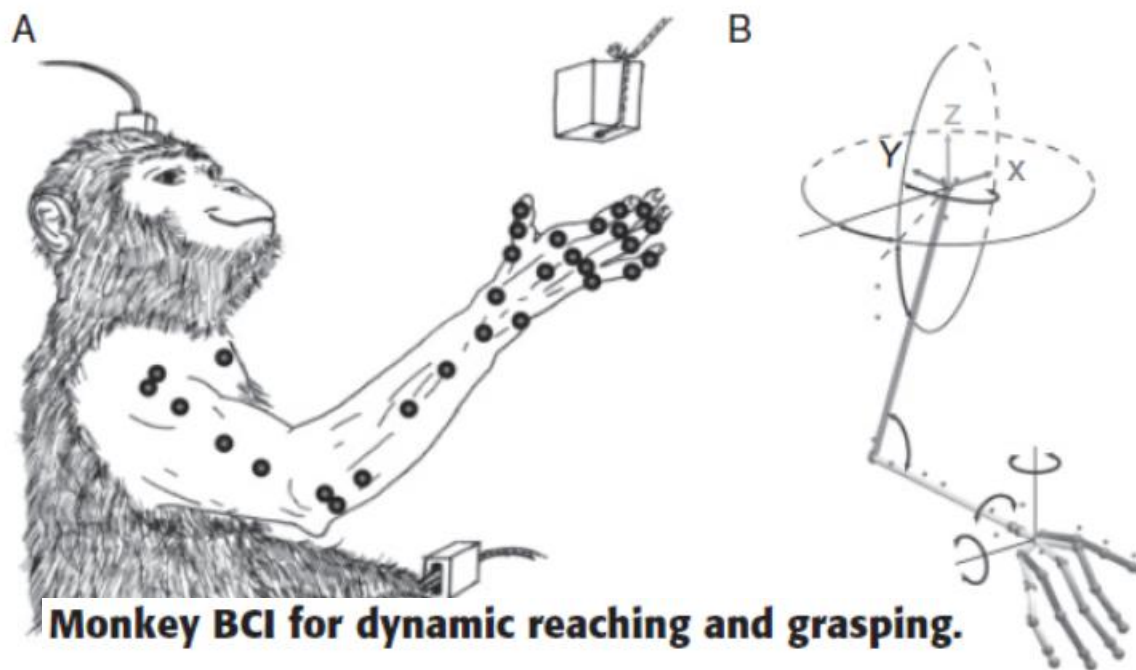




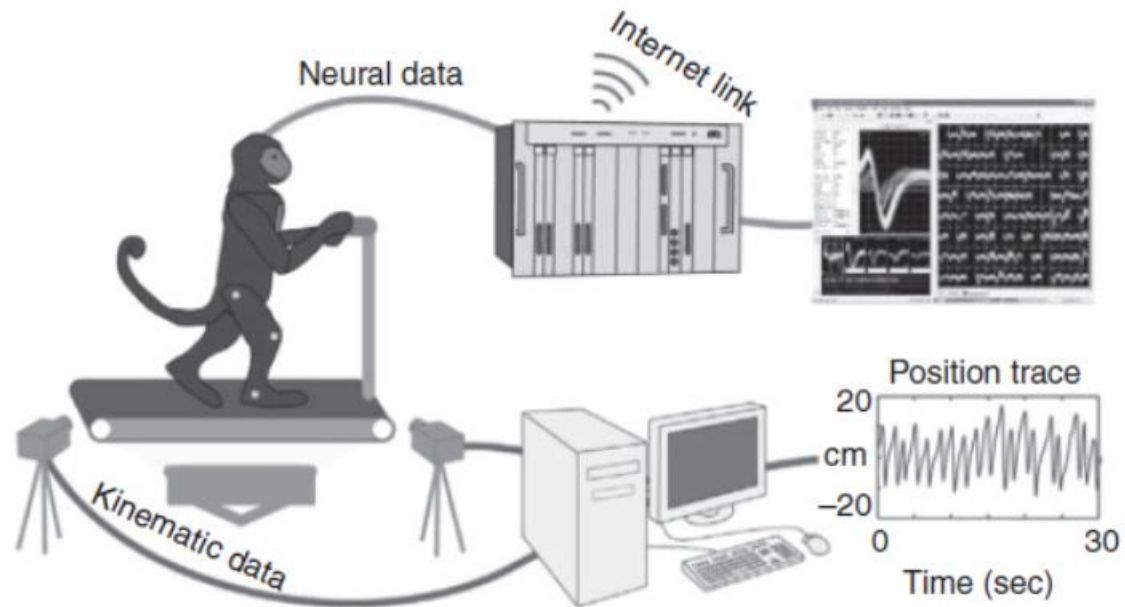
Monkey BCI for a one-dimensional control task.



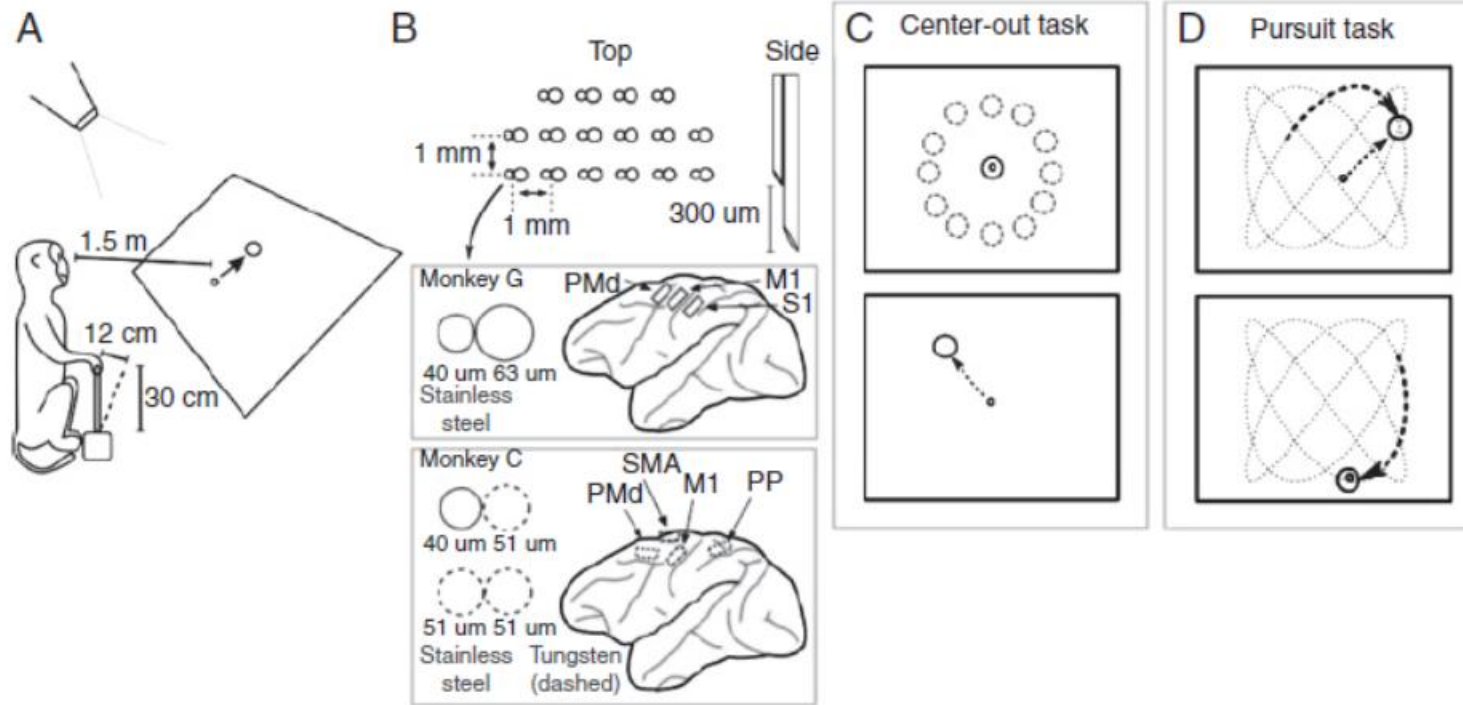
BCI control of a prosthetic arm and gripper for self-feeding.



A

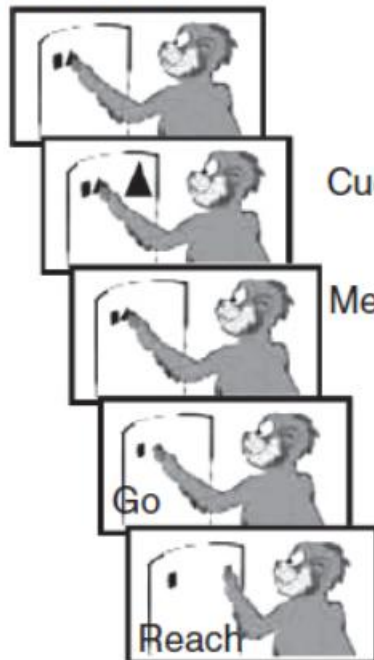


Predicting lower-limb kinematics using neural activity.

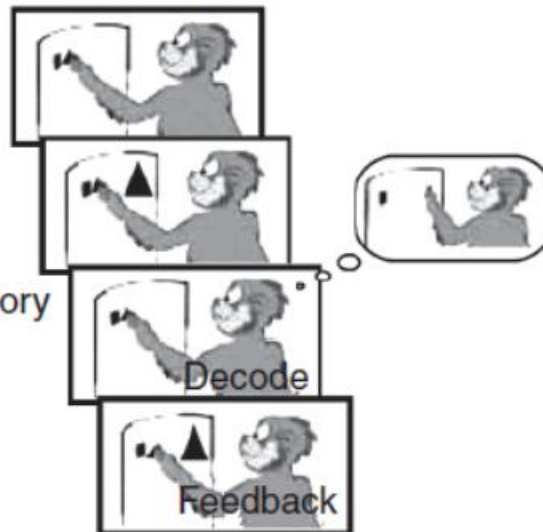


Experimental setup for demonstrating BCI control of a cursor.

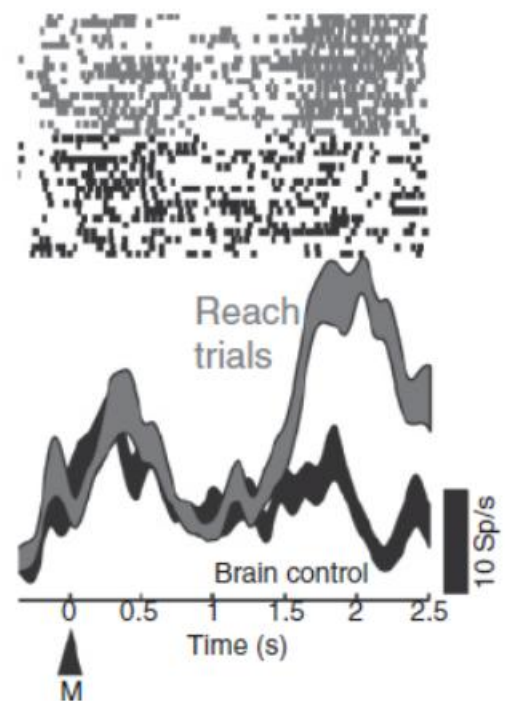
A Reach trial



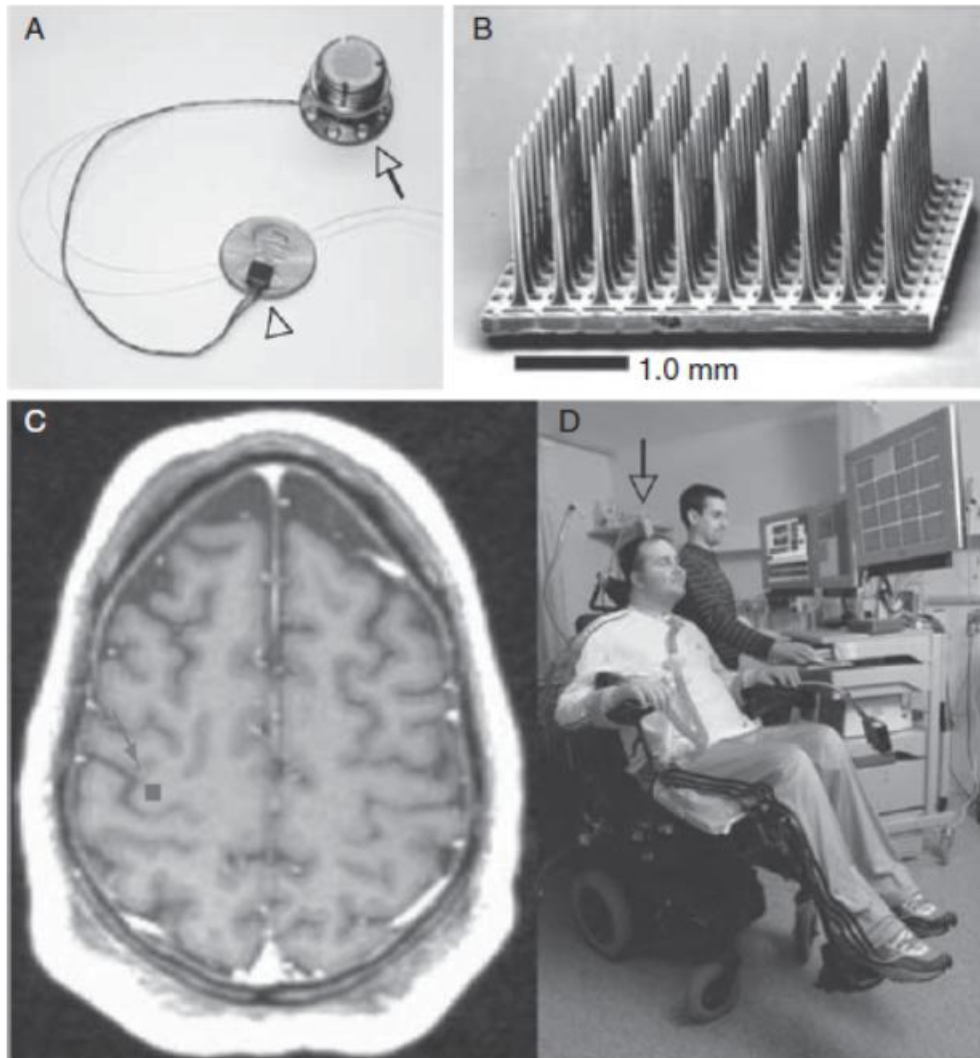
Brain control trial



B



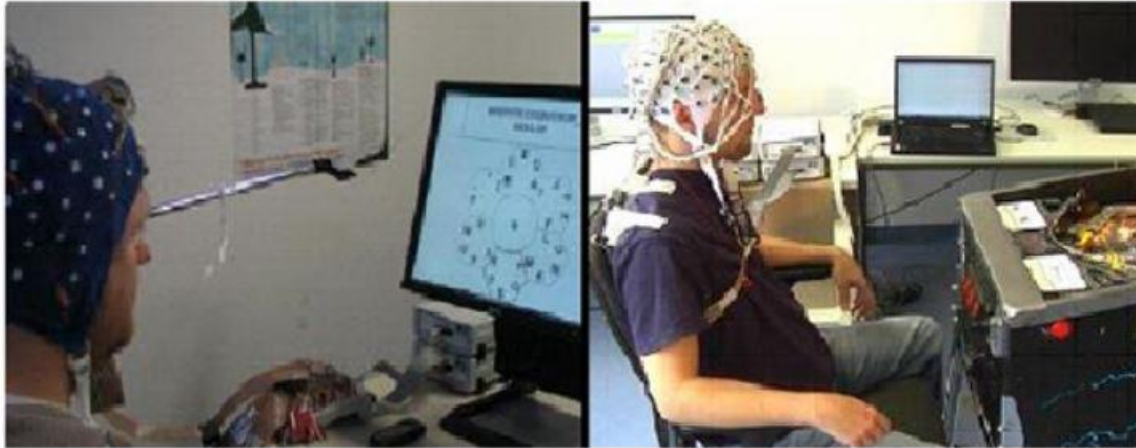
Cognitive BCI for a reaching task.



Invasive BCI in a human.

Minghui Shi, Xiamen University, China

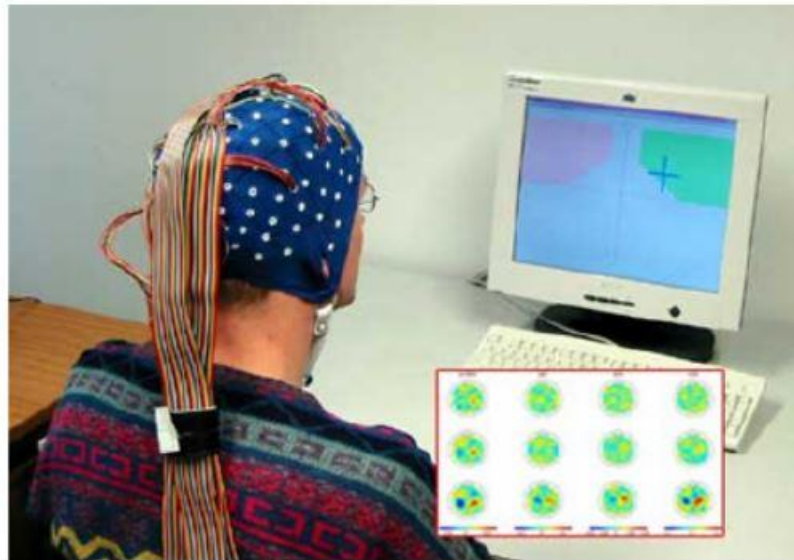
Figures of EEG-based BCIs



Berlin BCI: BCI Speller



Graz BCI: BCI for rehabilitation and virtual reality



Minghui Shi, Xiamen University, China



BCI for controlling robotic dogs



BCI based dialing system



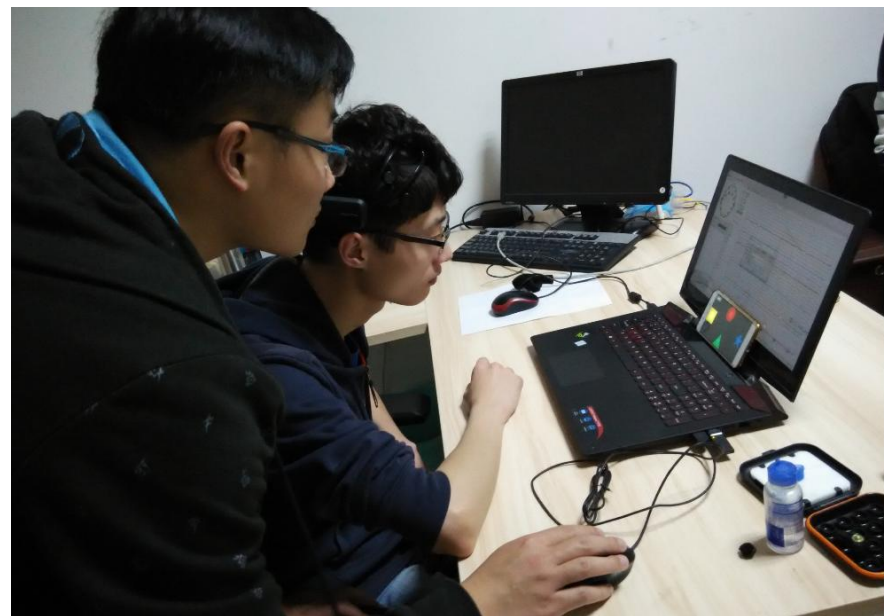
BCI for cursor controlling

BCI for wheelchair controlling



Minghui Shi, Xiamen University, China

Experiments in our lab



BCI Videos

- http://open.163.com/movie/2009/11/V/A/M6I12OHDE_M6I134KVA.html
- http://www.soku.com/search_video/q_brain%20computer%20interface?f=1&kb=0402000000000000_brain%20computer%20interface&rp=1451568674644XzYhjl&rp=1451568674644XzYhjl
- <http://www.youku.com/> search: brain computer interface
- http://v.youku.com/v_show/id_XMjMwMTQwNjQ0.html?from=s1.8-1-1.2
- http://v.youku.com/v_show/id_XMTM4NzE2MjAwMA==.html?from=s1.8-1-1.2
- http://v.youku.com/v_show/id_XODA0NTAyMjA4.html?from=s1.8-1-1.2
- http://v.youku.com/v_show/id_XMTM4NzE5Njg0MA==.html?from=s1.8-1-1.2
- [Our experiment 1](#) [Our experiment 2](#)

Summary



- Principle of Brain Computer Interface
- Content of this textbook
- Picture & Video

Thanks for your attention!